

Benchmarking Mixture of Experts and Fine Tuning Pretrained LLMs for Bio-Medical Multi-Class Classification

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Abstract

Current methodologies for biomedical text classification often utilize outdated Bi-Directional LSTM, machine learning algorithms, prompt engineering, or vanilla BERT encoders, failing to leverage recent breakthroughs in neural architectural designs. This research proposes a benchmark study comparing traditional domain-specific transformers against emerging architectural paradigms: Mixture of Experts (MoE) for sparse, efficient parameter scaling. By benchmarking these architectures against state-of-the-art encoder only models like BioBERT and BiomedBERT, we aim to determine if sparse expert routing can improve performance on classification task. Our evaluation focuses on optimizing the trade-off between modern expressivity and computational overhead, using Precision, Recall, F1-score and AUROC as the primary metrics to establish a new performance baseline for domain specific medical text analysis.

Keywords

Multi-class Classification, Fine-Tuning, Mixture-of-Experts

1 Introduction

Medical text classification is an important task in biomedical natural language processing, where the goal of the task is to automatically assign one or more relevant labels to abstracts of research papers, clinical notes, or scientific articles. However, this task can be particularly challenging due to the complexity of medical language, and the presence of domain-specific terminology. Medical text classification is important because it allows healthcare or research organizations to efficiently organize, retrieve, and analyze biomedical literature on a large-scale, which is crucial for researchers, clinicians, and healthcare workers who rely on obtaining relevant information in their field.

In our project, we wanted to benchmark BERT models that are pretrained on domain specific data against newer transformer architecture for medical text classification task. Our approach involves comparing the performance of models such as BiomedBERT, and a transformer model with Mixture of Experts (MoE) implemented, against the performance of our baseline model, which is BioBERT.

The contributions of our paper are as follows:

- We present a benchmark study comparing domain-specific pretrained transformer models (BioBERT and BiomedBERT) with a custom Mixture-of-Experts (MoE) transformer architecture for biomedical multi-class text classification.
- We design and implement a scalable Top-k Mixture-of-Experts transformer from scratch, incorporating expert routing, load balancing loss, and shared expert layers to improve parameter efficiency while maintaining model expressivity.

2 Problem Formulation

The medical text classification task we are trying to solve in this paper is a multi-class multi-class classification task. Given the biomedical text which consists of title, keywords, and abstract as input, the goal is to automatically assign one or more labels from these seven categories: Case Report, Diagnosis, Epidemic Forecasting, Mechanism, Prevention, Transmission, and Treatment.

There are a few challenges that make this task complicated. First, using biomedical text means that the specialized terminology used in these texts may not be well represented in models that are pretrained on data from general domain instead of a specific domain. Other than that, the dataset is imbalanced as certain categories such as Epidemic Forecasting appear far less frequently in the dataset than any other categories, which can cause the model’s performance to drop significantly when predicting the Epidemic Forecasting category.

We selected BioBERT as our baseline model, a domain-specific language model developed by Lee et al[7]. It is initialized from the original BERT architecture and further pretrained on large-scale biomedical corpora, including PubMed abstracts and PubMed Central full-text articles. BioBERT has established itself as a strong and widely adopted baseline for a broad range of biomedical NLP tasks, including named entity recognition, relation extraction, and text classification[7], making it relevant for evaluating the relative performance of newer architectures. In our implementation, we froze all encoder weights and trained only the classification head, which consists of a linear layer followed by a ReLU activation, dropout, and a final linear layer mapping to the seven output labels. This configuration allows us to isolate the quality of BioBERT’s pretrained representations without any task-specific adaptation.

3 Related Work

Many techniques used by papers who attempted this problem use outdated methodology. ML-Net: multi-class classification of biomedical texts with deep neural networks[4] attempted this problem by building a model using Bi-Directional LSTM and RNN layers. Although RNNs have shown to be good at processing NLP, it isn’t competitive to modern Transformer architectures.

AstroConcepts: A Large-Scale Multi-class Classification Corpus for Astrophysics[1] tries to solve a similar problem but for the astro-physics domain. They approached this problem by using Rule-Based Matching, K-Nearest Neighbors, Transform learning using specialized BERT models, and Vocabulary-Constrained LLMs. Rule-Based matching assumes that if a label appears in the title, then the paper belongs to that label. This is often true, however it fails to capture secondary labels that may correspond to the paper. K-Nearest Neighbors, using Qwen3-Embedding-8B, was used to

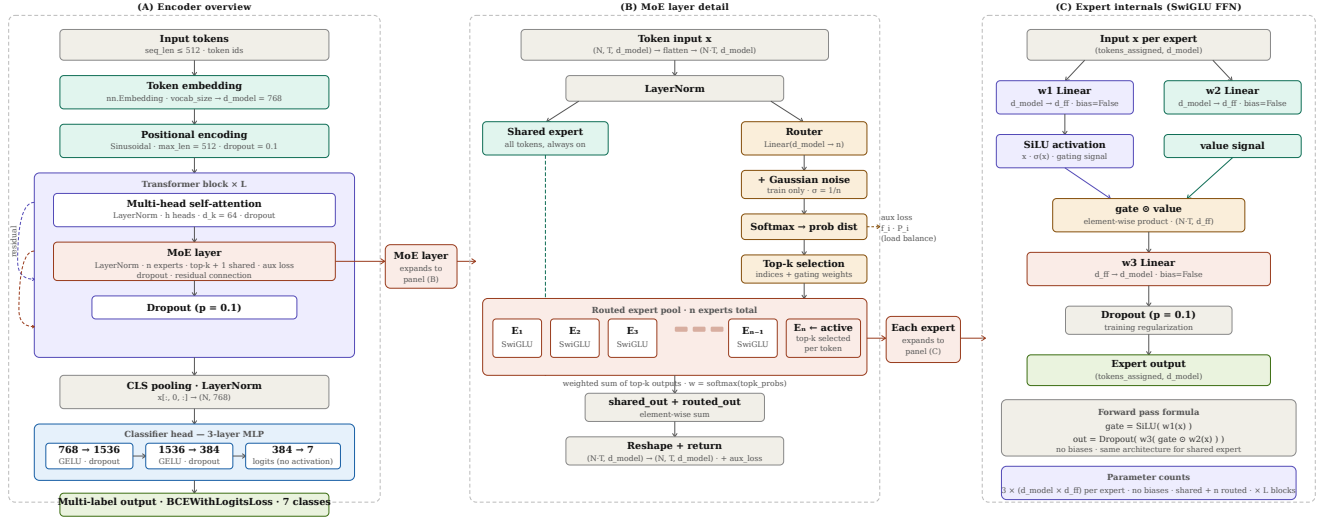


Figure 1: Overview of the MoE Encoder Architecture

retrieve the embeddings to compare the similarity using cosine similarity. Vocabulary-Constrained LLMs used astroBERT (their best performing model) to extract the top 50 labels from their dataset. Researchers then used DeepSeekV3-Reasoner and prompt engineering to prompt the LLM to select the best corresponding label. The researchers did attempt to use a transformer-based transfer learning. However both of these transformer architectures are quite outdated and new advancements in the transformer architecture have been found since their inception.

Cost-Aware Model Selection for Text Classification: Multi Objective Trade-offs Between Fine-Tuned Encoders and LLM Prompting in Production[5] also tried to approach this problem by finetuning LLMs and engineer prompting to achieve the best score. However, similar to the AstroConcepts: A Large-Scale Multi-class Classification Corpus for Astrophysics paper[1], this method falls short as it is dependent on old BERT models.

However, new efforts within the community have recently focused on leveraging modern advancements to address the broader implications of this issue. Recent research has moved away from general-purpose models in favor of domain-specific architectures. These models are developed using specialized datasets—most notably those trained on biomedical corpora from PubMed and Electronic Health Records [2]. Training on domain-specific corpora enables models to internalize medical terminology and context, leading to superior classification metrics compared to general-purpose baselines.

4 Model Architecture

4.1 Mixture of Experts

Our implementation of the Mixture of Experts layer utilizes the Top-k variation, a technique originally proposed by Google[8] and further improved upon by Deepseek[3]. While the naive MoE architecture used a fixed top-2 routing mechanism, subsequent research demonstrated that the number of active experts, k , can be optimized

as a tunable hyperparameter to balance computational efficiency with model expressivity.

Following DeepSeek’s MoE framework, our architecture integrates a shared expert component alongside a sparse routed expert pool. In this configuration, every token is processed by a commonly shared expert to capture universal patterns, while the router selectively activates a subset of specialized experts for greater granular feature extraction.

The routing process begins by flattening the input tokens from (N, T, d_{model}) to $(N * T, d_{\text{model}})$. The selection logic is determined by a linear router that projects the input into a 32-dimensional vector space. During the training phase, Gaussian noise is introduced to the router to encourage expert diversity. This discourages the router from selecting the same expert for every token. The expert selection is determined by the following steps:

- **Probability Distribution:** A Softmax function is applied to the router’s output to generate a probability distribution over the N available experts. In our implementation, we had 32 experts.
- **Top-k Selection:** For our specific implementation, we set $k = 3$, identifying the 3 experts with the highest activation weights for each token.
- **Weighted Aggregation:** The output is computed as a weighted sum of the top 3 active experts, which is then combined with the output from the shared expert.
- **Load Balancing:** To ensure efficient utilization of the expert pool, we incorporated an auxiliary loss $(f_i \cdot p_i)$ that penalizes unbalanced expert selection.

Each of the experts in the routed pool utilizes a SwiGLU architecture to maximize non-linear expressivity. This design utilizes two linear layers— w_1 and w_2 —to project the input x into a higher dimensional space. The forward pass within each expert is defined as:

$$\text{Output} = \text{Dropout}(w_3(\text{SiLU}(w_1(x)) \odot w_2(x)))$$

Model	Learnable Layers	Macro F1	Micro F1	Total Parameters	Active Parameters
BioBERT	Classification head	0.69	0.82	108,508,935	198,663
BiomedBERT	Classification Head	0.83	0.89	109,680,903	198,663
BiomedBERT	Last 1 + Classification Head	0.83	0.89	109,680,903	7,286,535
BiomedBERT	Last 2 + Classification head	0.85	0.89	109,680,903	14,374,407
MoE	All	0.74	0.80	978,617,863	14,155,776

Table 1: Model Performance Comparison

By omitting biases in these linear projections, the resulting architectural design ensures that each expert maintains a minimal parameter footprint. This efficiency allows for significant model scaling without a proportional increase in inference latency.

4.2 Finetuning BiomedBERT

For our finetuned BiomedBERT model, it is a BERT-based encoder model that is pretrained on full biomedical abstracts and full text articles from PubMed[6]. Instead of training all the layers in the BiomedBERT model, we used a different way of finetuning in order to reduce computational cost and overcome the issue of the model overfitting in early epochs. Our best finetuned BiomedBERT model has all its encoder weights frozen except for the last two transformer layers, so that the model could learn from our dataset and adapt to our specific task. Different finetuning strategy was also tested, specifically, we also tested by freezing all the encoder weights and only training the classifier head, and freezing all encoder weights except for the last transformer layer.

On top of freezing most of the encoder weights, we also used a custom classification head with a linear layer of 256 neurons, followed by ReLU activation function, then a dropout layer of 0.3, and a final linear layer of 7 neurons to produce predictions for our 7 target labels. During the forward pass, the [CLS] token representation from the final hidden state was extracted and passed through this classification head to produce the output logits.

The model was trained using Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss), which internally applies a sigmoid activation before computing the loss and is well-suited for our multi-class multi-class classification task. We used the AdamW optimizer with a learning rate of $2e-5$ and applied gradient clipping to prevent exploding gradients during back propagation. The input text was tokenized using BiomedBERT's corresponding tokenizer, with a maximum sequence length of 256 tokens. The model with the best performance was trained for 10 epochs with a batch size of 16.

5 Experimental Evaluation

5.1 Methodology

Our dataset can be found at the following link: <https://ftp.ncbi.nlm.nih.gov/pub/lu/LitCovid/biocreative/>. In particular, we utilized BC7-LitCovid-Train.csv for our training dataset and BC7-LitCovid-Test-GS.csv as our test dataset. The original dataset consisted of the following features: pmid, journal, title, abstract, keywords, pub_type, authors, doi, and labels. However for our implementation we only included title, keywords and abstract (in this particular order). We

decided to structure our data in this order to ensure that important classification data from the title and keywords wouldn't be truncated if the tokens were longer than 512 words. All null values were replaced with an empty string. To speed up our training times, we decided to create a small subset of the dataset. Our train set had 10,000 rows while our test had 2,000, resulting in a 80/20 test train split. The following metrics were used to compare our model's performance:

- Precision
- Recall
- F1
- AUROC

For our implementation of MoE, special tokens like [CLS] and [SEP] were manually added. We found that the model performed better if the tokens were manually rather than having the model automatically add them during the embedding process. Training for our MoE was done on a H200 through LightningAi. Given that we had 141 GB of VRAM to use, we decided to maximize the size of our model to utilize the massive compute capability that we had access to. This resulted in us using the following hyperparameters for the MoE model:

- max_len=512
- d_k=64
- d_model=768
- num_heads=12
- dropout_prob=0.1
- n_layers=8
- num_experts=32
- top_k=3
- batch_size=128

5.2 Results

Table 1 shows the performance of all models evaluated in this benchmark. Among the finetuned BERT-based models, BiomedBERT consistently outperformed the baseline BioBERT model across all finetuning configurations. BioBERT, trained with only the classification head unfrozen, achieved a Macro F1 of 0.69 and a Micro F1 of 0.82, suggesting that its pretrained weights did not transfer as effectively to our specific task without unfreezing more transformer layers. In contrast, BiomedBERT with the same classifier-only finetuning strategy already achieved a Macro F1 of 0.83 and a Micro F1 of 0.89, highlighting the advantage of pretraining on a corpora more similar to our dataset.

Furthermore, unfreezing additional transformer layers in BiomedBERT yielded further improvements. Allowing the last two transformer layers to be trainable alongside the classification head produced our best overall result, achieving a Macro F1 of 0.85 and a Micro F1 of 0.89, with 14,374,407 active parameters out of roughly 109 million total. Notably, unfreezing only the last one transformer

Labels	Per-label Precision	Per-label Recall	Per-label F1	AUROC
Case Report	0.9016	0.6707	0.7692	0.9688
Diagnosis	0.7311	0.7923	0.7605	0.9201
Epidemic Forecasting	0.6538	0.5000	0.5667	0.9642
Mechanism	0.7924	0.7667	0.7794	0.9535
Prevention	0.8289	0.9127	0.8688	0.9669
Transmission	0.5743	0.5524	0.5631	0.9330
Treatment	0.7800	0.9020	0.8366	0.9305

Table 2: MoE Performance Metrics

layer produced identical Macro and Micro F1 scores to the classifier-only configuration. This indicates a point of diminishing returns where the added computational cost of unfreezing more layers may not always translate into proportional performance gains. In our case, doubling the active parameters in BiomedBERT did not lead to much improvements in Macro F1 and Micro F1 scores.

The custom MoE model that we build achieved a lower Macro F1 score of 0.7349, underperforming compared to all BiomedBERT variants. While the MoE architecture significantly increases the total number of parameters to 978 million, it maintains a relatively low number of active parameters of 14 million, due to sparse expert routing. This highlights the trade-off between computational cost and model performance, although the MoE model is computationally efficient during inference, it does not necessarily translate to better performance in our domain-specific classification task. We think that MoE architectures hold promise for biomedical classification tasks, though further tuning of the routing mechanism and training procedure may be needed to close the performance gap with encoders pretrained on domain-specific data.

6 Conclusion

In this paper, we presented a benchmark study comparing domain-specific pretrained models against a custom Mixture of Experts architecture for biomedical multi-class multi-class text classification on the BC7-LitCovid dataset. Our results showed that BiomedBERT, pretrained on full biomedical abstracts and full-text articles from PubMed, consistently outperformed BioBERT across all finetuning configurations, with our best BiomedBERT model achieving a Macro F1 of 0.85 when the last two transformer layers and classification head were left trainable. This result highlights the importance of pretraining on domain-specific data, as BiomedBERT's stronger representations allowed it to adapt more effectively to our classification task even with a minimal number of active parameters.

Our custom MoE architecture, while achieving a lower Macro F1 of 0.7349, demonstrated a higher parameter efficiency. With total parameters of roughly 978 million but only approximately 14 million active parameters during inference, the MoE model was able to approach BioBERT-level performance, validating the premise that sparse expert routing can scale model capacity without proportional increases in computational cost. However, the performance gap between the MoE model and the best BiomedBERT configuration suggests that architectural innovations alone are not sufficient to overcome the advantages brought by domain-specific pretraining. Future work could explore combining both approaches, for example,

by pretraining an MoE architecture directly on biomedical corpora, or by incorporating more sophisticated routing strategies to improve expert specialization and overall classification performance.

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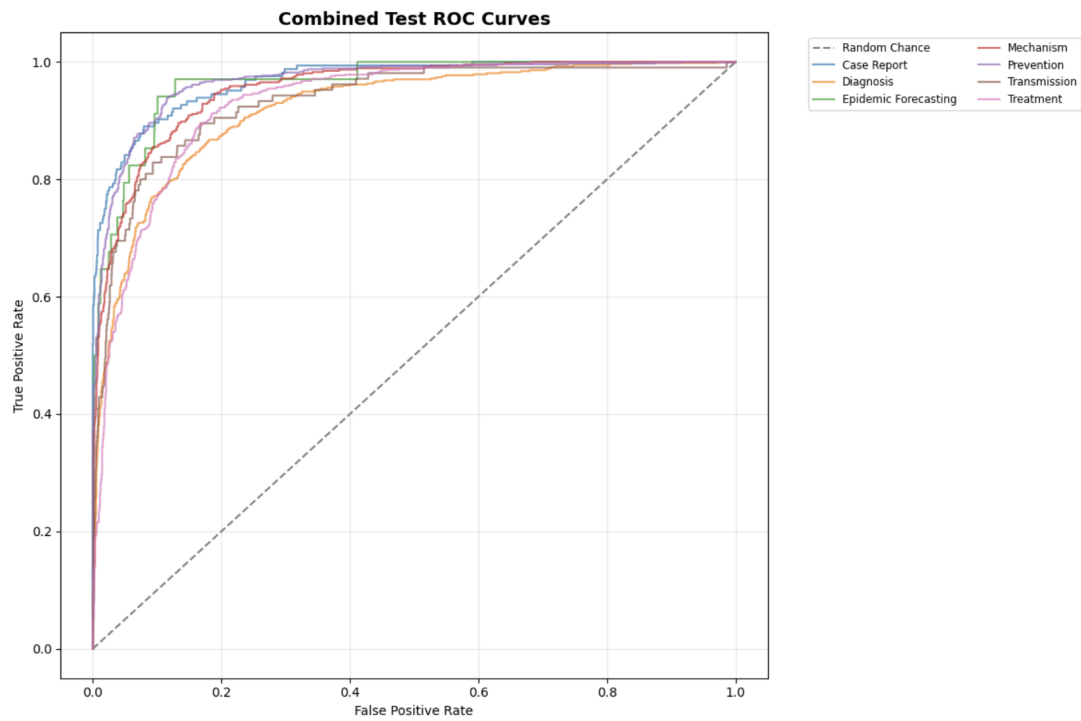


Figure 2: MoE AUROC Plot

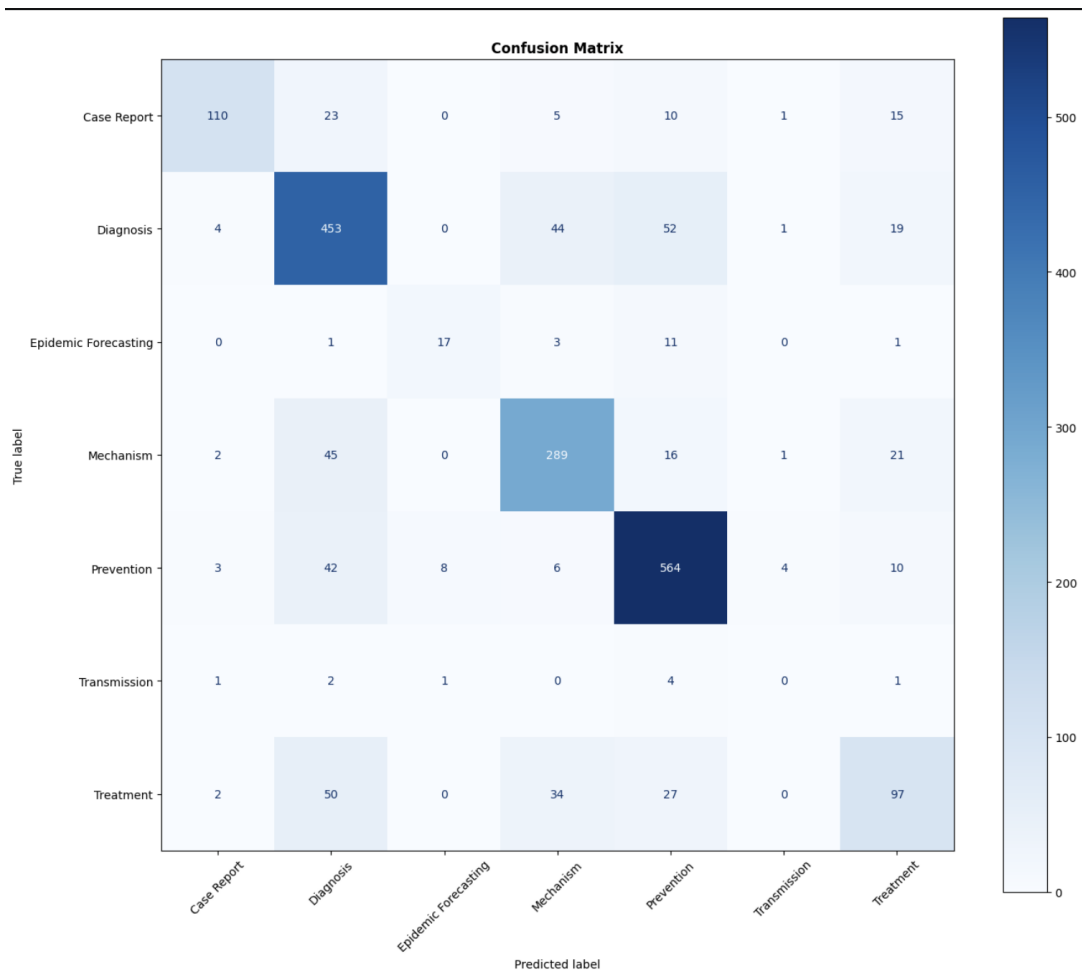


Figure 3: MoE Confusion Matrix

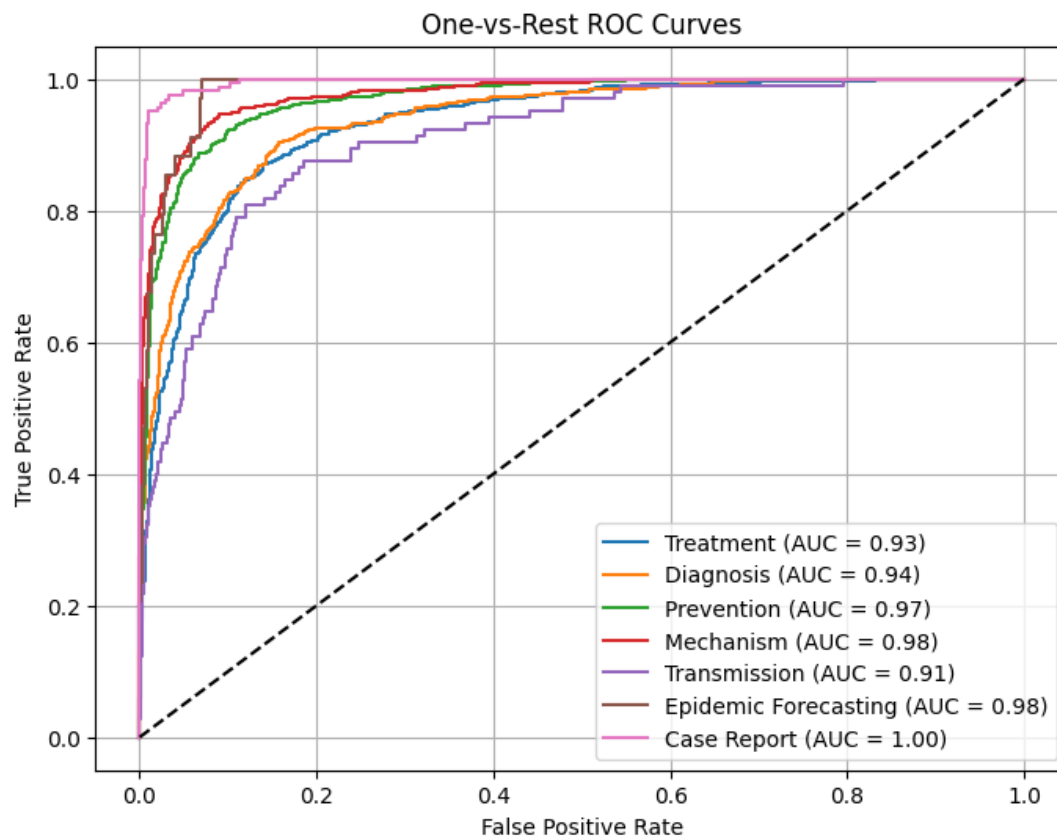


Figure 4: BiomedBERT with classification head AUROC Plot

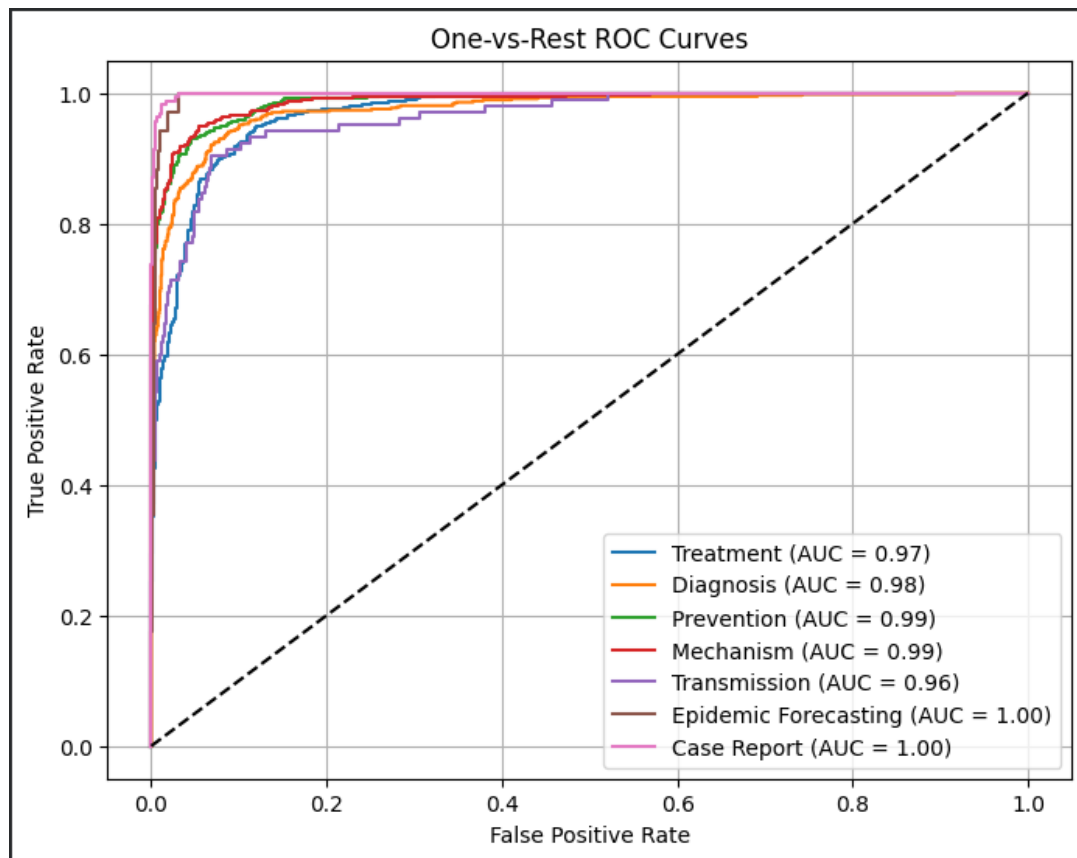


Figure 5: BiomedBERT with last layer and classification head AUROC Plot

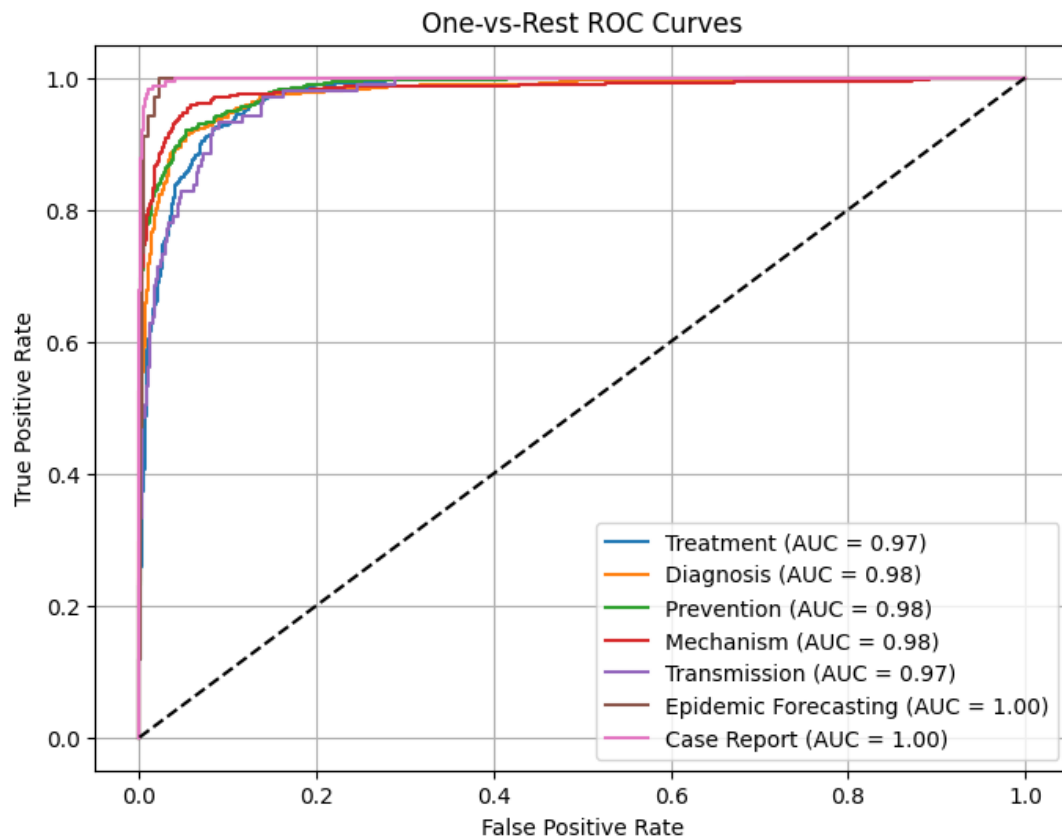


Figure 6: BiomedBERT with last 2 layers and classification head AUROC Plot

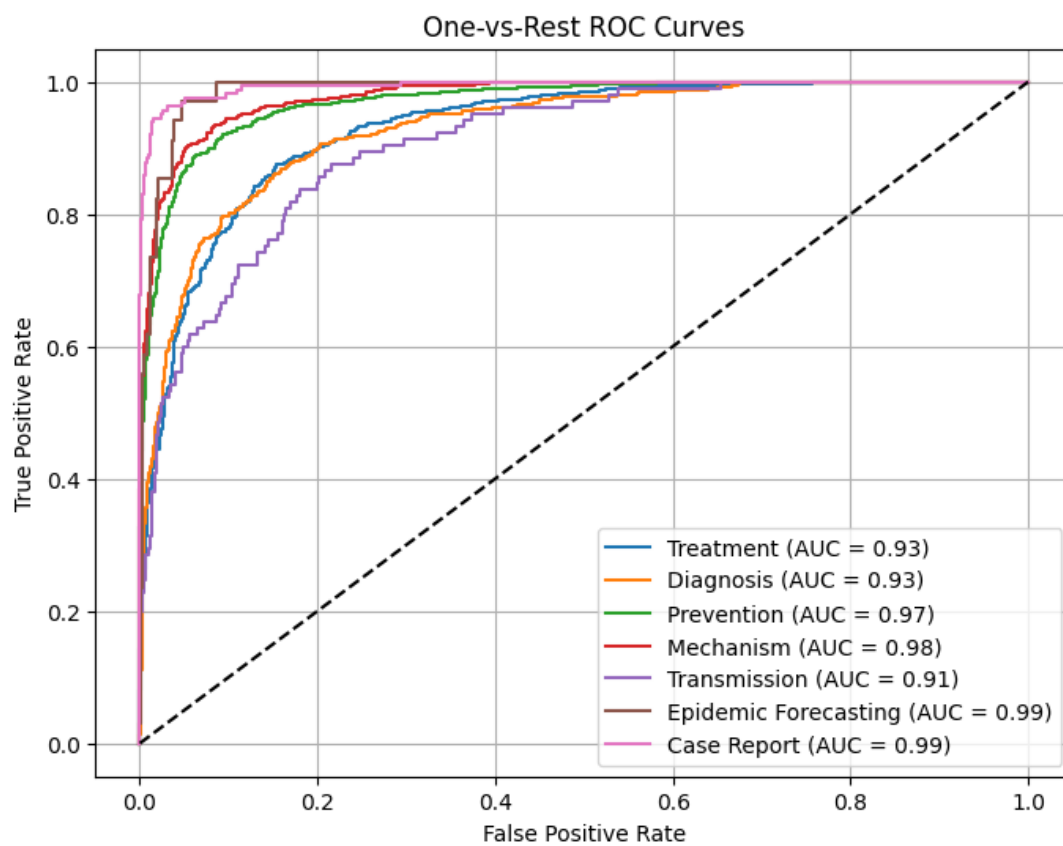


Figure 7: BioBERT with classification head AUROC Plot